MATH 161, Autumn 2025
SCRIPT 3: Introducing a Continuum

This sheet introduces a continuum C through a series of axioms.

Axiom 1. *A continuum is a nonempty set C.*

We often refer to elements of C as *points*.

Definition 3.1. Let X be a set. An *ordering* on the set X is a subset $<$ of $X \times X$, with elements $(x, y) \in <$ written as $x < y$, satisfying the following properties:

(a) *(Trichotomy)*

For all $x, y \in X$ exactly one of the following holds: $x < y$, $y < x$ or $x = y$.

(b) *(Transitivity)* For all $x, y, z \in X$, if $x < y$ and $y < z$ then $x < z$.

Remarks 3.2. a) In mathematics "or" is understood to be inclusive unless stated otherwise. So in a) above, the word "exactly" is needed.

b) $x < y$ may also be written as $y > x$.

c) By $x \leq y$, we mean $x < y$ or $x = y$; similarly for $x \geq y$.

Axiom 2. *A continuum C has an ordering $<$.*

Definition 3.3. If $A \subset C$ is a subset of C, then a point $a \in A$ is a *first* point of A if, for every element $x \in A$, either $a < x$ or $a = x$. Similarly, a point $b \in A$ is called a *last* point of A if, for every $x \in A$, either $x < b$ or $x = b$.

Lemma 3.4. *If A is a nonempty, finite subset of a continuum C, then A has a first and last point.*

Proof. We want to prove that if $A \subset C$, and A is finite and nonempty, then A has a first and a last point. We want to prove this through induction.

If $|A| = 1$, then $A = \{a\}$ for some $a \in C$, and therefore we may trivially say a is both the least and greatest element of A. Therefore, the base case is true.

Let A be a subset where $|A| = k + 1$. We may choose some $a \in A$, and define $B = A \setminus \{a\}$. Then B is nonempty and finite with $|B| = k$, so by the inductive hypothesis, B has a least element m.

As these are subsets of the continuum, they have an ordering $<$. Therefore, either $a \leq m$ or $m \leq a$.

Case 1: If $a \leq m$, then for all $x \in A$ we have $a \leq m \leq x$, so a is the least element of A.
Case 2: If $m \leq a$, as m is the least element of B, then $m \leq x$ for all $x \in A$. Hence m is the least element of A. Therefore, we may say A has a least element.

For the greatest element, consider the reversed order $>$ on C. By the symmetry argument (or replacing $<$ with $>$) we may show that A must also have a greatest element.

$\square$

Theorem 3.5. *Suppose that A is a set of n distinct points in a continuum C, or, in other words, $A \subset C$ has cardinality n. Then symbols $a_1, \ldots, a_n$ may be assigned to each point of A so that $a_1 < a_2 < \cdots < a_n$, i.e. $a_i < a_{i+1}$ for $1 \leq i \leq n-1$.*

Proof. We want to prove the theorem by induction.

To prove the base case, let $n = 1$. Therefore, $|n| = 1$ and $A = \{a_1\}$, which does not need an ordering as it is a single element.

Suppose $n = k$, where $k > 1$. Assume that $A \subset C$ with $|A| = k + 1$. By **Lemma 3.4**, we may say that A has a first point a and a last point b. In this case, let us say the first point is a_1 and the last point is a_{k+1}. Therefore, if we remove the last point of A, a_{k+1}, then we get the set A' where $A' = A \setminus \{a_{k+1}\}$ with a cardinality of k. By induction hypothesis, A' may be written as $a_1 < a_2 < \cdots < a_k$. As we have removed the last element from A to get A' with an ordering $<$, the union set of A' and a_{k+1} would equal the set A.

$$A = A' \cup a_{k+1}$$

As we know A' is ordered as $a_1 < a_2 < \cdots < a_k$ and a_{k+1} is the last element, the ordering of A must be $a_1 < a_2 < \cdots < a_k < a_{k+1}$. Therefore, this works to prove the theorem.

□

Definition 3.6. If $x, y, z \in C$ and either (i) both $x < y$ and $y < z$ or (ii) both $z < y$ and $y < x$, then we say that y is *between* x and z.

Corollary 3.7. *Of three distinct points in a continuum, one must be between the other two.*

Proof. We want to show that if we have three distinct points in a continuum, then one of the points must be between the other two. Let $x, y, z \in C$ and suppose none of them are equal.

Suppose $A \subset C$ in which $A = \{x, y, z\}$. As $A \subset C$, by **Lemma 3.4**, it has a first and last point. Therefore, suppose x is the first point of A and z is the last point of A, then y must be between the two points. Therefore, of the three points in a continuum, as one must be the first point and one must be the last point, the third point must be between.

Hence, we have proven that if there are three distinct points in a continuum, one of them must be between the other two.

□

Axiom 3. *A continuum C has no first or last point.*

In the next exercise we show that the integers and the rationals both have orderings that satisfy Axioms 1-3.

Exercise 3.8. a) We define a relation $<$ on $\mathbb{Z}$ by $m < n$ if $n = m + c$ for some $c \in \mathbb{N}$. Show that, $\mathbb{Z}$, with the ordering $<$, satisfies Axioms 1-3.

Proof of a. Axiom 1: We want to prove that $\mathbb{Z}$ is nonempty. We know that $\mathbb{Z}$ contains 20 as an element. Therefore, trivially, as $\mathbb{Z}$ contains an element, it is not nonempty. This proves axiom 1.

Axiom 2: Suppose $x, y \in \mathbb{Z}$. If $x = y$, the statement holds trivially. If $x \neq y$, then $x - y$ is a nonzero integer, so either $x - y > 0$ or $x - y < 0$.

Case 1: $x - y > 0$ then $x > y$ must be true.
Case 2: $x - y < 0$ then $x < y$ must be true.

Therefore, we have shown that only one of the relations of $x > y, \;\; x < y, \;\; x = y$ is true. Therefore, trichotomy is true.

To prove transitivity, suppose $x, y, z \in \mathbb{Z}$ and they are all distinct points. Given that they are three different points, according to **Corollary 3.7**, one is between the other two. Therefore, we may say $x < y$ and $y < z$, where y is between the other two. This implies $x < z$, which proves transitivity.

Axiom 3: We want to prove that there is no first or last element in $\mathbb{Z}$.

Firstly, suppose k is the last element in $\mathbb{Z}$. As $\mathbb{Z}$ is an infinite set, we know that there is some variable $k + 1 \in \mathbb{Z}$, where $k + 1 > k$. As there exists an element greater than the arbitrary element k, there is no last element in $\mathbb{Z}$

Secondly, suppose $k \in \mathbb{Z}$. As $\mathbb{Z}$ is an infinite set, there exists some element $k - 1$ such that $k - 1 < k$. Therefore, as there is always an element that is less than the arbitrary element k, there is no first element in $\mathbb{Z}$. Hence, this works to prove Axiom 3.

□

b) Show that, for any $p = \left[\frac{a}{b}\right] \in \mathbb{Q}$, there is some $(a_1, b_1) \in p$ with $0 < b_1$.

Proof. Since $b \in \mathbb{Z}$ and $b \neq 0$, b must be either positive ($b > 0$) or negative ($b < 0$).

Case 1: $0 < b$: We may choose the representative pair $(a_1, b_1) = (a, b)$.Since $0 < b$, we have $0 < b_1$.The pair (a, b) is trivially in the equivalence class p.

Case 2: $b < 0$: If b is negative, let $a_1 = -a$ and $b_1 = -b$.Since $b < 0$, multiplying by -1 reverses the inequality, so $b_1 = -b > 0$.We must now verify that (a_1, b_1) is equivalent to (a, b):

$$a_1 b = b_1 a$$

Substituting the definitions of a_1 and b_1:

$$(-a)(b) = (-b)(a)$$

$$-ab = -ab$$

This equality holds. Thus, $(a_1, b_1) \sim (a, b)$, and the pair $(a_1, b_1) \in p$ satisfies $0 < b_1$. □

c) We define a relation $<_{\mathbb{Q}}$ on $\mathbb{Q}$ as follows. For $p, q \in \mathbb{Q}$, let $(a_1, b_1) \in p$ be such that $0 < b_1$, and let $(a_2, b_2) \in q$ be such that $0 < b_2$. Then we define $p <_{\mathbb{Q}} q$ if $a_1 b_2 < a_2 b_1$. Show that $<_{\mathbb{Q}}$ is a well-defined relation on $\mathbb{Q}$.

Proof. We want to show that if $\left[\frac{a}{b}\right] = \left[\frac{a'}{b'}\right]$ and $\left[\frac{c}{d}\right] = \left[\frac{c'}{d'}\right]$ with $b, b', d'd' > 0$ and $ad < bc$, then $a'd' < b'c'$.

We know that $ab' = a'b$ and $cd' = c'd$. Hence, $ab'dd' = a'bdd'$, where we are multiplying by dd' since $d, d' > 0$.

Then, $ab'dd = abdd' < bcb'd'$. Therefore, $ab'dd' < bcb'd'$, hence $ab'dd'cd'bb'$, hence$ab'dd' < c'dbb'$. Hence $a'd'bd < b'v'bd$, hence $a'd' < b'c'$. This completes the proof.

□

d) Show that $\mathbb{Q}$, with the ordering $<_{\mathbb{Q}}$, satisfies Axioms 1-3.

Proof. Axiom 1: We know that $\left[\frac{1}{2}\right] \in \mathbb{Q}$, and therefore $\mathbb{Q} \neq \emptyset$.

Axiom 2: We want to show that $\mathbb{Q}$ holds the principle of transitivity when it has an ordering of $<$.

To prove this we want to show that if $\left[\frac{a}{b}\right] < \left[\frac{c}{d}\right]$ and $\left[\frac{c}{d}\right] < \left[\frac{e}{f}\right]$ then $\left[\frac{a}{b}\right] < \left[\frac{e}{f}\right]$.

If $ad < bc$ and $cd < de$, then we may say $adf < bcf$ and $bcf < bde$. Hence, $bcf < bde$. Therefore, we may say $af < be$. Therefore, we have proven Axiom 2.

Axiom 3: We know that $\mathbb{Q}$ includes all positive and negative values, and therefore, we may say that it has no first or last point. Hence, we have proven Axiom 3. □

Definition 3.9. If $a, b \in C$ and $a < b$, then the set of points between a and b is called a *region*, denoted by $\underline{ab}$.

Warning 3.10. One often sees the notation (a, b) for regions. We will reserve the notation (a, b) for ordered pairs in a product $A \times B$. These are very different things.

Theorem 3.11. *If x is a point of a continuum C, then there exists a region $\underline{ab}$ such that $x \in \underline{ab}$.*

Proof. Suppose $x \in C$, where $x \neq a$ and $x \neq b$. Let $A \subset C$ such that $a, b, x \in A$. By **Lemma 3.4**, we say that there must be a first and last point between a, b, and x. Suppose a is the first point and b is the last point. Through **Corollary 3.7** we may say that if there are three distinct points in a continuum, then there must be one element between the other two. Therefore, we may say that $x > a$ and $x < b$. Hence, there exists a region $\underline{ab}$ such that $x \in \underline{ab}$.

□

We now come to one of the most important definitions of this course:

Definition 3.12. Let A be a subset of a continuum C. A point p of C is called a *limit point* of A if every region R containing p has nonempty intersection with $A \setminus \{p\}$. Explicitly, this means:

$$\text{for every region } R \text{ with } p \in R, \text{ we have } R \cap (A \setminus \{p\}) \neq \emptyset.$$

Notice that we do not require that a limit point p of A be an element of A. We will use the notation $LP(A)$ to denote the set of limit points of A.

Theorem 3.13. *If $A \subset B$, then $LP(A) \subset LP(B)$.*

Proof. We want to prove that if $A \subset B$ we may therefore state that $LP(A) \subset LP(B)$.

Suppose $x \in LP(A)$. By the definition of a limit point, for every region R containing x there exists a point $p_a \in R \cap (A \setminus \{x\}$. Similarly, as $A \subset B$, every point in A will be a point in B. Therefore, we have a point $p_b \in R \cap (B \setminus \{x\})$. Hence, we can say $x \in LP(B)$. As $A \subset B$, we may say $R \cap (A \setminus \{x\} \subset R \cap (B \setminus \{x\}$. This can be written as $LP(A) \subset LP(B)$. Hence, we have proven the theorem. □

Definition 3.14. If $\underline{ab}$ is a region in a continuum C, then $C \setminus (\{a\} \cup \underline{ab} \cup \{b\})$ is called the *exterior* of $\underline{ab}$ and is denoted by $\operatorname{ext} \underline{ab}$.

Lemma 3.15. *If $\underline{ab}$ is a region in a continuum C, then*

$$\operatorname{ext} \underline{ab} = \{x \in C \mid x < a\} \cup \{x \in C \mid b < x\}.$$

Proof. We want to prove the existence of the exterior $\underline{ab}$ through double inclusion.

Firstly, let $x \in C$ either $x < a$ or $x > b$. Therefore, x is not between a and b and $x \notin \{a, b\}$. Hence, x lies in the exterior $\underline{ab}$ so $x \notin \{a, b\} \cup \underline{ab}$ Thus we may write .

$$x \in C \setminus (\underline{ab} \cup \{a, b\})$$

.

Therefore, every x is either $x < a$ or $x > b$ and therefore belongs to exterior of $\underline{ab}$. Hence, we may write

$$\{x \in C \mid x < a\} \cup \{x \in C \mid b < x\} \subseteq \text{ ext } \underline{ab}$$

Secondly, let $x \in \text{ext } \underline{ab}$. Therefore, $x \notin ab \cup \{a, b\}$. We may say that, by trichotomy, exactly one of $x < a$, $a \leq x \leq b$, or $b < x$ holds true. Then, as $x \notin \{a\}$ and $x \notin \{b\}$, as we may say $x < a$ or $x > b$. Therefore, we may write this as $\{x \in C | x < a\} \cup \{x \in C | b < x\}$. Further, we can state that $\text{ext}\underline{ab} \subseteq \{x \in C | x < a\} \cup \{x \in C | b < x\}$.

Therefore, through double inclusion, we have proven the lemma.

□

Lemma 3.16. *No point in the exterior of a region is a limit point of that region. No point of a region is a limit point of the exterior of that region.*

Proof. Suppose first that $x \in \text{ext}(ab)$. Then $x < a$ or $x > b$.

Case 1: $x < a$

Let $c \in C$ where $c < x < a$. The interval ca is an open segment containing x and lying entirely to the left of ab. Therefore, we may say that $ca \cap ab = \emptyset$ and $ca \cap (ab \setminus \{x\}) = \emptyset$. Therefore, we may say that x cannot be a limit point of ab. Hence $x \notin LP(ab)$.

Case 2: $x > b$.

Let $c \in C$ where $b < x < c$. The interval bc contains x and lies to the right of ab. Therefore, we may say that $bc \cap ab = \emptyset$ and $bc \cap (ab \setminus \{x\}) = \emptyset$. Therefore, we may again say that x cannot be a limit point of ab and hence $x \notin LP(ab)$.

Through these two cases, we have shown that no exterior point is a limit point of the region.

Now suppose $x \in ab$. Then $x \notin \text{ext}(ab)$, and removing x from the exterior leaves it unchanged:

$$\text{ext}(ab) \setminus \{x\} = \text{ext}(ab).$$

Since the region and its exterior are disjoint,

$$ab \cap (\text{ext}(ab) \setminus \{x\}) = \emptyset.$$

Thus x is not a limit point of $\text{ext}(ab)$.

$\square$

Theorem 3.17. *If two regions have a point x in common, their intersection is a region containing x.*

Proof of Theorem 3.17. We want to prove that if two regions have a common point, their intersection is a region that contains the common point.

Let $a, b, c, d \in C$ such that there exists regions $\underline{ab} \in C$ and $\underline{cd} \in C$. Suppose there exists a common element x such that $x \in \underline{ab}$ and $x \in \underline{cd}$. Therefore, by **Definition 3.9**, if $x \in \underline{ab}$ then $a < x < b$ and if $x \in \underline{cd}$ then $c < x < d$.

Now, we know that the continuum C has an ordering $<$, and hence there must exists a $\max\{a, c\}$ and a $\min\{c, d\}$. Suppose there exists points $e, f \in C$ such that $e = \max\{a, c\}$ and $f = \min\{c, d\}$. Therefore, we may say that there exists a region $\underline{ef} \in C$ such that $\underline{ef} = \underline{ab} \cap \underline{cd}$. As $x \in \underline{ab}$ and $x \in \underline{cd}$, $\max\{a, c\} < x < \min\{b, d\}$. Hence, $x \in \underline{ef}$. Therefore, we have proven the statement.

$\square$

Corollary 3.18. *If n regions $R_1, \ldots, R_n$ have a point x in common, then their intersection $R_1 \cap \cdots \cap R_n$ is a region containing x.*

Proof of Corollary 3.18. We want to prove that if a point x is common in n regions, then their intersection will contain it as well. We will prove this through induction.

Suppose we have n regions $R_1, \ldots, R_n$. Firstly, we want to prove the base case. Let the base case be $n = 2$. Therefore, there exist regions $R_!$ and R_2 that have a point x in common. By **Theorem 3.17**, we may say that if $x \in R_1$ and $x \in R_2$, then $x \in R_1 \cap R_2$.

Now, as the base case is true let us assume that the statement is true for n regions $R_1, R_2, \ldots, R_n$. Then, we want to show that if there exists a region R_{n+1} such that $x \in R_{n+1}$, then the intersection $R_1 \cap R_2 \ldots R_n \cap R_{n+1}$ is a region containing x.

Through the inductive process we assume $x \in R_1 \cap R_2 \ldots \cap R_n$. Now, consider $R_1 \cap R_2 \ldots \cap R_n$ as the set R' such that

$$R' = R_1 \cap R_2 \ldots \cap R_n$$

As $x \in R_1 \cap R_2 \ldots \cap R_n$, we may say that $x \in R'$. Now, let there be an intersection $R' \cap R_{n+1}$. By **Theorem 3.17**, we may say that $x \in R' \cap R_{n+1}$. By the definition fo R' we may write that

$$x \in R' \cap R_{n+1}$$

$$x \in (R_1 \cap R_2 \ldots \cap R_n) \cap R_{n+1}$$

$$x \in R_1 \cap R_2 \ldots \cap R_n \cap R_{n+1}$$

Thus, as this statement holds true for all values of n, this works to prove that if n regions have a point x in common, then their intersection is a region containing x. □

Theorem 3.19. *Let A, B be subsets of a continuum C. Then $LP(A \cup B) = LP(A) \cup LP(B)$.*

Proof of Theorem 3.19. We want to prove that $LP(A \cup B) = LP(A) \cup LP(B)$ through double inclusion.

Firstly, we want to show that $LP(A) \cup LP(B) \subset LP(A \cup B)$. By **Theorem 1.7** we can state that $A \subset A \cup B$ and $B \subset A \cup B$. By **Theorem 3.13** we can write the expressions as $LP(A) \subset LP(A \cup B)$ and $LP(B) \subset LP(A \cup B)$. Hence, we can say that $LP(A) \cup LP(B) \subset LP(A \cup B)$.

Secondly, we want to show that $LP(A \cup B) \subset LP(A) \cup LP(B)$. We may do this through a contradiction. For the sake of contradiction, suppose that there exists a point p such that $p \in LP(A \cup B)$ and $p \notin LP(A) \cup LP(B)$.

Therefore, we can say that $p \notin LP(A)$ and $p \notin LP(B)$. Hence, there exists a region R_A containing p such that $R_A \cap (A \setminus \{p\}) = \emptyset$ and there exists a region R_B containing p such that $R_B \cap (B \setminus \{p\}) = \emptyset$.

Let there be a region R such that $R = R_A \cap R_B$. By **Theorem 3.17**, if both regions R_A and R_B contain p, then their intersection R must also contain p. Thus R is not empty as it contains at least the point p. Therefore, to determine whether R contains any limit points, we examine the expression

$$R \cap ((A \cup B) \setminus \{p\}) = [R \cap (A \setminus \{p\})] \cup [R \cap (B \setminus \{p\})]$$

We see that as the RHS is empty, the LHS is empty as well. The equation therefore leads to

$$R \cap ((A \cup B) \setminus \{p\}) = \emptyset \cup \emptyset = \emptyset$$

However, this contradicts the assumption that $p \in LP(A \cup B)$. Therefore, our assumption that $p \notin LP(A)$ and $p \notin LP(B)$ must be false. Hence, if $p \in LP(A \cup B)$, then $p \in LP(A)$ or $p \in LP(B)$. This shows that $LP(A \cup B) \subset LP(A) \cup LP(B)$, completing the double inclusion.

□

Corollary 3.20. *Let $A_1, \ldots, A_n$ be n subsets of a continuum C. Then p is a limit point of $(A_1 \cup \cdots \cup A_n)$ if, and only if, p is a limit point of at least one of the sets A_k.*

Proof of Corollary 3.20. We want to prove that p is a limit point of the union of n sets if at least one of the sets contains the limit point.

Let $A_1, \ldots, A_n$ be n subsets of a continuum C. Firstly, we need to prove that if p is a limit point of one set. Let $p \in LP(A_k)$ and $\forall \lambda \in n, p \notin A_\lambda$. By **Theorem 3.19**, we know that $LP(A \cup B) = LP(A) \cup LP(B)$. As $LP(A \cup B)$ becomes a new set, this can contiune as $LP(A_1 \cup A_2 \cup A_3 \cup A_4) = LP(A_1 \cup A_2) \cup LP(A_3 \cup A_4)$. Through this, we can take the union of all n subsets of C. Hence, even if $\forall \lambda \in n, p \notin A_\lambda$, there exists a A_k such that $p \in A_k$. Therefore we may say that

$$LP(\bigcup_{\forall \lambda \in n} A_\lambda) = LP(A_1) \cup LP(A_2),\ \ldots, \cup LP(A_k) \cup, \ldots \cup LP(A_n)$$

and as $p \in A_k$ we can say that $p \in LP(\bigcup_{\lambda \in n} A_\lambda)$.

Secondly, we need to show that this applies if more than one subset contains the limit point. We may use the same principle as if there was one set containing p. If $p \in A_{k1}$ and $p \in A_{k2}$, then it is still present in $\bigcup_{\lambda \in n} A_\lambda$ as it only requires one set to contain it. If $\forall A_\lambda, p \notin A_\lambda$ then it is not present in the union set as at least one set must contain the limit point.

Hence, this works to prove that p is a limit point of $(A_1 \cup \cdots \cup A_n)$ if, and only if, p is a limit point of at least one of the sets A_k. □

Theorem 3.21. *If p and q are distinct points of a continuum C, then there exist disjoint regions R and S containing p and q, respectively.*

Proof of Theorem 3.21. . Let p and q be points in the continuum C. Suppose that there exists two regions S and R such that $p \in S$ and $q \in R$. We want to show that $q \notin R \cap (A \setminus \{p\})$.

Let R have the first and last points of a and b respectively and S have the first and last points of c and d respectively. By **Definition 3.9**, we know that as $p \in R$, $a < p < b$

□

Corollary 3.22. *A subset of a continuum C consisting of one point has no limit points.*

Proof of 3.22. We want to show that if a subset of a continuum has only one point, it does not contain any limit points.

Let $A \subset C$. We know that $A = \{a\}$ and $|A| = 1$. By **Definition 3.12**, a limit point p of A exists when for every R where $p \in R$, we have a nonempty intersection with $A \setminus \{p\}$. As $p \in A$ and $|A| = 1$, it must be that $p = a$. Furthermore, as the A contains only one point, there does not exists any region where it can contain other points of the set besides the point itself. Therefore, the set $A \setminus \{p\} = A \setminus \{a\} = \emptyset$ and therefore for all regions $R \cap (A \setminus \{p\} \neq \emptyset$. Hence, if a set contains only one point it does not contain any limit points. □

Theorem 3.23. *A finite subset A of a continuum C has no limit points.*

Proof of Theorem 3.23. We want to prove that a finite subset of a continuum has no limit points. We will show this through direct proof.

Let $A \subset C$ and suppose A is finite. A point $p \in C$ is a limit point of A if every region R containing p has a nonempty intersection with $A \setminus \{p\}$. Therefore, we need to show that there exists such a region that $R \cap A \setminus \{p\} = \emptyset$. As A is finite, we may say that it has a set cardinality n and therefore has a confined number of elements.

Case 1: $p \in A$. If $p \in A$, it must be an element of A. Since $A \subset C$, the elements of A can be listed in increasing order. Let a_{i-1} and a_{i+1} be the nearest points of $p \in A$ to the left and right of p. Suppose there exists points $a, b \in C$ such that

$$a_{i-1} < a < p < b < a_{i+1}.$$

Therefore, there exists a region $\underline{ab}$ such that $p \in \underline{ab}$ and contains no other points of A. Thus, we may say that $\underline{ab} \cap (A \setminus \{p\}) = \emptyset$ as the only point of intersection between $\underline{ab}$ and A is p. Therefore, this case contains no limit points.

Case 2: $p \notin A$. Let $p \notin A$. Furthermore, a_1 be the first element of A and let a_n be the last element of A. Now, we may say that $p < a_1$ or $p > a_n$.

Firstly, assume $p < a_1$. By the definition of C we may say that exists $c, d \in C$ such that $c < p < d < a_1$. Hence, we may say there exists a region $\underline{cd}$ where $p \in \underline{cd}$ and $p \notin A$. As $\underline{cd}$ contains no points in A, we may say that $\underline{cd} \cap A \setminus \{p\} = \emptyset$. Thus, it contains no limit points.

Secondly, assume $p > a_n$. By the definition of C we may say that exists $e, f \in C$ such that $a_n < e < p < f$. Hence, we may say there exists a region $\underline{ef}$ where $p \in \underline{ef}$ and $p \notin A$. As $\underline{ef}$ contains no points in A, we may say that $\underline{ef} \cap A \setminus \{p\} = \emptyset$. Thus, it contains no limit points.

This completes the proof. □

Corollary 3.24. *If A is a finite subset of a continuum C and $x \in A$, then there exists a region R, containing x, such that $A \cap R = \{x\}$.*

Proof of Corollary 3.24. We want to prove that if a finite set of a continuum contain an element, there must be a region that contains x.

Let A be a finite subset of the continuum C. Suppose $A = \{a, x, b\}$ and $|A| = 3$. As the set is finite we know that there contains no limit points as well as there exists a first and a last point. Let a be the first point and b be the last point and $a < x < b$. As A is finite, there exists no limit points. Hence, $R \cap (A \setminus \{x\}) = \emptyset$. Hence, as x is the only element contained in both R and in A, we may therefore say that $A \cap R = \{x\}$ as A must contain only the elements a, b, x. Therefore, there exists a region R that contains x such that $A \cap R = \{x\}$ □

Theorem 3.25. *If p is a limit point of A and R is a region containing p, then the set $R \cap A$ is infinite.*

Proof of Theorem 3.25. We want to prove this by contradiction.

Let p be a limit point of A and region R contain p and $R \cap A \neq \emptyset$. Suppose that $R \cap A$ is not infinite. By **Theorem 3.23** we know that $R \cap A$ cannot therefore contain a limit point p as finite sets do not contain limit points. Hence, there must be a region R' such that $R' \cap A$ contain limit points. To contain limits they must not be finite to ensure that there exists some region such that $R' \cap (A \setminus \{p\}) \neq \emptyset$

Therefore, we may say that if there exists limit points for A and there exists a region R containing p then the set $R \cap A$ requires the existence of limit points. Hence, as the set cannot contain limit points and be finite, it must be that $R \cap A$ is infinite. □

Exercise 3.26. Find realizations of a continuum $(C, <)$. That is, find concrete sets C endowed with a relation $<$ satisfying all of the axioms (so far). Are they the same? What does "the same" mean here?

Additional Exercises

In all exercises you are expected to prove your answer, unless explicitly stated otherwise.

1. Show that for all $p, q \in \mathbb{Q}$ such that $p <_{\mathbb{Q}} q$, there is some $r \in \mathbb{Q}$ with $p <_{\mathbb{Q}} r <_{\mathbb{Q}} q$.
2. By identifying $n \in \mathbb{Z}$ with $[\frac{n}{1}] \in \mathbb{Q}$ we can think of $\mathbb{Z}$ as a subset of $\mathbb{Q}$. We shall write n to mean $[\frac{n}{1}]$. Show that for all $p \in \mathbb{Q}$, there is some $n \in \mathbb{Z}$ such that $p <_{\mathbb{Q}} n$.
3.

   Definition 3.27. Given two sets A and B, we say that $A \subsetneq B$ if $A \subset B$ and $A \neq B$.

   For each set X and subset $<_X \subset X \times X$, determine if $<_X$ is an ordering:

   (a) Let $X = \wp(\mathbb{N})$ and $<_X = \{(A, B) \in \wp(\mathbb{N}) \times \wp(\mathbb{N}) | A \subset B\}$.
   (b) Let $X = \big\{\{x \in \mathbb{N} | x \leq n\} \in \wp(\mathbb{N}) | n \in \mathbb{N}\big\}$ and $<_X = \{(A, B) \in X \times X | A \subsetneq B\}$.

(c) Let $X = \{f \subset \mathbb{N} \times \mathbb{N} | f \text{ is a function}\}$ and $<_X = \{(f, g) \in X \times X | f(n) < g(n) \text{ for all } n \in \mathbb{N}\}$.

(d) Let $X = \{f \subset \mathbb{N} \times \mathbb{N} | f \text{ is a function}\}$ and $<_X = \{(f, g) \in X \times X | f(n) \leq g(n) \text{ for all } n \in \mathbb{N} \text{ and there exists } n \in \mathbb{N} \text{ such that } f(n) < g(n)\}$.

(For the purposes of this exercise, "$<$" means the usual ordering on $\mathbb{N}$, i.e. if $m, n \in \mathbb{N}$ then $m < n$ if and only if $n = m + k$ for some $k \in \mathbb{N}$.)

4. Which of the following pairs of a set and an ordering satisfy Axioms 1,2, and 3 (and are thus examples of a "continuum")?

   i) $C_1 = \{[n] \in \mathcal{P}(\mathbb{N}) \mid n \in \mathbb{N} \cup \{0\}\}$, where, for $[n], [m] \in C_1$, we say $[n] <_{C_1} [m]$ if $[n] \subsetneq [m]$;
   ii) $C_2 = \mathbb{Z}$, where $<_{\mathbb{Z}}$ is the usual ordering on $\mathbb{Z}$;
   iii) $C_3 = \mathbb{Z} \times \mathbb{N}$, where, for $(a, b), (x, y) \in \mathbb{Z} \times \mathbb{N}$ we say that $(a, b) <_3 (x, y)$ if $a < x$ or if $a = x$ and $b < y$.

5. Find, without proof, the exterior of each region.

   (a) Let C_2 and $<_{\mathbb{Z}}$ be as in Exercise 5. Let $R = \underline{38}$.
   (b) Let C_3 and $<_3$ be as in Exercise 5. Let $R = \underline{(-2, 5)(-2, 10)}$.
   (c) Let C_3 and $<_3$ be as in Exercise 5. Let $R = \underline{(-2, 5)(1, 10)}$.

6. Find the limit points of each region.

   (a) Let C_2 and $<_{\mathbb{Z}}$ be as in Exercise 5. Let $R = \underline{38}$.
   (b) Let C_3 and $<_3$ be as in Exercise 5. Let $R = \underline{(-2, 5)(-2, 10)}$.
   (c) Let C_3 and $<_3$ be as in Exercise 5. Let $R = \underline{(-2, 5)(1, 10)}$.

7. Let A and B be realizations of the continuum, with orderings $<_A, <_B$, respectively. We say that A and B are *isomorphic* if there is a bijection $f : A \longrightarrow B$ such that

$$a_1 <_A a_2 \Longrightarrow f(a_1) <_B f(a_2).$$

   Show that $\mathbb{Z}$ and $\mathbb{Q}$ (with the orderings given in Scripts 1 and 2) are realizations of the continuum that are *not* isomorphic.

8. Suppose that $R_1, R_2, \ldots$ are regions in a continuum C and that $x \in LP(\bigcup_{i=1}^{\infty} R_i)$. Is it true that there exists $i \in \mathbb{N}$ such that $x \in LP(R_i)$? Does this answer change if we add the additional hypothesis that there exists $k \in \mathbb{N}$ such that $x \in R_k$?

9. Show that any continuum C satisfying Axioms 1-3 is infinite and write C as a union of regions in C.

10. Prove that if $A \subset \mathbb{Z}$, A has no limit points.

11. Let $i^2 = -1$ and define $\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\}$. Define an ordering on $\mathbb{Z}[i]$ by $a + bi < c + di$ if, and only if, either $a < c$, or $a = c$ and $b < d$.

(a) Prove that $<$ is indeed an ordering. (So you must verify the properties in Definition 3.1. Note that you may assume the usual properties of $\mathbb{Z}$ - see Script 0 for full details.)

(b) Prove that, with this ordering, $\mathbb{Z}[i]$ satisfies Axioms 1,2 and 3, so is a realization of the continuum as defined so far.